RESEARCH ARTICLE

SituationalLLM: Proactive Language Models with Scene Awareness for Dynamic, Contextual Task Guidance

[version 1; peer review: 2 approved with reservations]

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Abstract

Background

Large Language Models (LLMs) have demonstrated remarkable success in text-based reasoning tasks but struggle to provide actionable guidance in real-world physical environments. This limitation arises from their lack of situational awareness—an inability to recognize gaps in their understanding of a user's physical context, leading to unreliable and overly generic instructions. To address this, we propose **SituationalLLM**, a novel approach that integrates structured scene representations into LLMs to improve context-aware assistance.

Methods

SituationalLLM leverages scene graphs—structured representations of objects, attributes, and spatial relationships—to encode real-world environments in a text-based Scene Graph Language. We introduce the **Situational Awareness Database for Instruct-Tuning (SAD-Instruct)**, which pairs diverse scene graphs with multi-agent dialogue, enabling LLMs to iteratively refine their guidance through clarifying questions. A **LoRA-adapted LLaMA-3-8B model** is fine-tuned on SADInstruct to bridge structured knowledge with natural language reasoning, enhancing its ability to recognize missing information and dynamically adjust responses.

Results

Qualitative evaluations show that SituationalLLM outperforms state-ofthe-art LLMs (GPT-4, LLaMA-3) in providing precise, task-specific,

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and contextually relevant instructions. The model reduces hallucinations by proactively identifying missing environmental details and requesting clarifications before generating guidance. Through comparative analyses on everyday tasks (e.g., cooking, office assistance), SituationalLLM demonstrates superior adaptability, delivering grounded, user-centered recommendations.

Conclusion

By integrating structured scene representations and iterative dialogue-based refinements, SituationalLLM enables more reliable, context-aware AI assistants. This research highlights the significance of bridging structured knowledge with natural language for enhanced real-world task guidance. Future work should focus on expanding scenario diversity and improving real-time scene perception to further enhance situational adaptability.

Keywords
Large Language Models, Scene Graphs, Context-Aware Assistance, Situational Awareness

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1 Introduction

Large language models (LLMs) have demonstrated exceptional capabilities in understanding and generating human language[1–3]. This has driven advancements in a wide range of tasks, ranging from language understanding[4,5] and software development[6] to human motion descriptions[7]. Recent work is increasingly focused on creating LLM-based AI systems that operate in real-world environments[8]. This includes both virtual assistants like chatbots and embodied agents like robots. However, traditional LLMs struggle to leverage accurate world models of their surroundings, limiting their effectiveness in physical contexts[9]. This is a significant barrier to developing robust LLM-driven systems that can help complete tasks in the real world.

Current LLMs often fail to recognize their incomplete understanding of the physical world and rarely seek clarifications[10]. Consequently, when used for task guidance, they tend to produce verbose and overly generic instructions with hallucinations that rely on broad assumptions, making them difficult for users to interpret and apply in their situation. For embodied agents, this can lead to ineffective or even unsafe task execution[11], highlighting the critical need for actionable and contextually accurate task guidance.

Figure 1 demonstrates this limitation with an example from GPT-4[3]. Here, the model assumes the jar is stubborn and provides broad, non-specific solutions without understanding the user's exact context (e.g., why help is needed, the jar's characteristics, etc.). In contrast, if the user asks another human for help with a similar task, that person would likely attempt to understand why the user needs assistance before providing a targeted response specific to the user's circumstances. The LLM's lack of awareness about the user's context and inability to recognize its own incomplete understanding, or what we term situational awareness[12,13], results in ill-suited guidance. This highlights the need for an LLM to recognize incomplete context and proactively seek details to refine its guidance.

The lack of detailed descriptions of the physical world in standard user interactions with LLMs makes it difficult for LLMs to understand the context. For LLM-driven virtual assistants to function more naturally and be user-friendly, instead of relying entirely on the user to provide all necessary context, a practical assistant should proactively identify missing details and ask clarifying questions, much like a human would. This ability to extract context through interaction would make systems more intuitive and reduce the burden on users to provide complete, precise information upfront. In the case of LLM-driven embodied agents, some environmental context can be gleaned through sensors attached to the agent. When combined with scene-understanding tools, these sensors can offer structured representations of the environment—scene graphs—which can then be added to prompts to improve situational awareness.

However, conventional LLMs are not inherently designed to process or reason with structured data like scene graphs. To address this gap, we propose a lightweight *Scene Graph Language* that encodes objects, their attributes, and the relationships between them as text. By converting each scene's data into a standardized textual form, the LLM can parse and reason about the environment more effectively. This format is particularly crucial in real-world settings where contextual details—such as the number of objects, their positions, or their relationships—impact task feasibility. Despite the potential of incorporating such structured information, there is a lack of standardized methods for integrating environmental inputs into LLM workflows. This makes producing actionable, context-aware responses tailored to users' needs challenging. In response, we introduce the **Situational Awareness Database for Instruct-Tuning (SAD-Instruct)**, a novel dataset that pairs scenario-specific scene graphs with multi-agent dialogue to generate context-sensitive instructions. SAD is derived from large-scale 3D semantic scene graphs[14,15], ensuring realistic and diverse indoor scenarios. Alongside scene-graph pruning and

I need help opening this glass jar on my desk.

Opening a stubborn glass jar can be tricky, but here are a few techniques that can help:

1. **Grip Enhancement:**
   - Use a rubber glove or a dish towel to increase grip on the lid.
   - Alternatively, wrap a thick rubber band around the lid for better traction.
2. **Tapping the Lid:**
   - Gently tap around the edges of the lid with the back of a knife or a wooden spoon. This may loosen the seal.
   - You can also tap the bottom of the jar against a soft surface to release pressure inside.
3. **Hot Water Trick:**
   - Run hot water over the lid (not the jar itself) for about 30 seconds. The metal expands faster than the glass, making it easier to twist.
4. **Leverage Tools:**
   - If you have a jar opener, it's perfect for this. If not, try using a spoon or a butter knife to pry a small gap under the lid to let air in.

Give one of these a try and let me know how it goes!

Figure 1. GPT-4 provides comprehensive but generic guidance when assisting with physical tasks, failing to account for specific user situations and constraints. It presumes that the jar is "stubborn" and neglects to ask for details like the type of jar or the user's limitations, which can lead to less applicable advice. *An ideal LLM-driven AI assistant should provide tailored advice, considering the user's real-world situation, implying a need for awareness of their physical context*.

multi-turn dialogue, SAD includes a wide range of tasks and reflection-style interactions designed to teach LLMs how to identify context gaps and ask clarifying questions.

To demonstrate the effectiveness of our dataset in eliciting desired behaviors in LLMs, we propose **SituationalLLM**, a model fine-tuned using LoRA[16] to effectively handle scene graph inputs, query missing details, and provide more grounded, user-centered guidance. Our method demonstrates the importance of bridging structured representations (scene graphs) with natural language dialogues, enabling LLMs to reason about what matters in a given situation. We present initial qualitative examples showing SituationalLLM outperforming GPT-4 and Llama 3 in situational awareness by dynamically adapting to user needs and leveraging environmental context. While these results are promising, they represent a step toward more robust, context-aware AI systems rather than a definitive solution.

1.1 Contributions

- We propose SituationalLLM, a fine-tuned language model designed to integrate structured scene knowledge for actionable, context-aware task guidance, benefiting both virtual assistants and embodied agents.
- We introduce the Situational Awareness Database for Instruct-Tuning (SAD-Instruct), a novel dataset that teaches LLMs to reason over structured and unstructured knowledge and refine guidance through iterative dialogue.
- We demonstrate, through qualitative examples, that SituationalLLM improves task specificity, reduces hallucinations, and provides grounded, reliable assistance, setting a foundation for AI systems capable of building and leveraging world models.

2 Methods

This section presents our approach for creating **SituationalLLM**, which provides actionable, context-aware task guidance by leveraging structured scene graphs and interactive multi-agent dialogue. We begin with a high-level overview of our solution, introduce the Scene Graph Language for structured representations, detail the construction of the Situational Awareness Database (SAD), and then describe how we fine-tune our model.

2.1 Overview of the proposed approach

Traditional LLMs struggle with executing tasks in the physical world because they rely primarily on text-based pretraining and often fail to incorporate crucial environmental details. Our proposed solution, **SituationalLLM**, addresses this challenge in three key steps:

1. **Encode environments as scene graphs:** We use the 3DSSG[14] dataset to obtain comprehensive semantic scene graphs, capturing objects, attributes, and relationships in real-world indoor environments (Section 2.2).
2. **Construct SAD with scenario-specific dialogues:** We generate situational contexts (scenarios) from these scene graphs and employ a multi-agent system to produce iterative dialogues and step-by-step instructions (Section 2.3).
3. **Fine-tune an LLM with scene graph awareness:** Finally, we train a LoRA adapter[16] on top of LLaMA-3-8b-Instruct[17] to integrate structured and unstructured knowledge (Section 2.4), yielding the *SituationalLLM* model.

2.2 Scene graph language and representation

Scene graphs offer structured, graph-based representations of environments, encoding objects (*nodes*), their attributes, and pairwise relationships (*edges*). To make such information accessible to an LLM, we propose converting it into a standardized **Scene Graph Language**:

```
obj-<label>-<id>:[<attr1>,<attr2>,...]; ...;
rel-<id>:(<subject>-<id>,<predicate>,<object>-<id>); ...;
```

Each `obj-<label>-<id>` entry details the object's name, ID, and any relevant attributes (e.g., "wooden," "large," "on table"). Relationship lines specify how objects interact (e.g., "under," "contains," "next to"). We can teach standard LLMs to parse and reason over environmental contexts without specialized architectures by translating scene graphs into this uniform textual format and fine-tuning lightweight adapters.

2.3 Constructing the Situational Awareness Database (SAD)

Our dataset, **SAD**, is built to teach LLMs situational awareness through structured scene graphs and multi-turn dialogues. Figure 2a illustrates the overall pipeline. Below, we detail how we generate diverse scenarios, create scenario-specific scene graphs, and employ a multi-agent system to produce high-quality instructions.

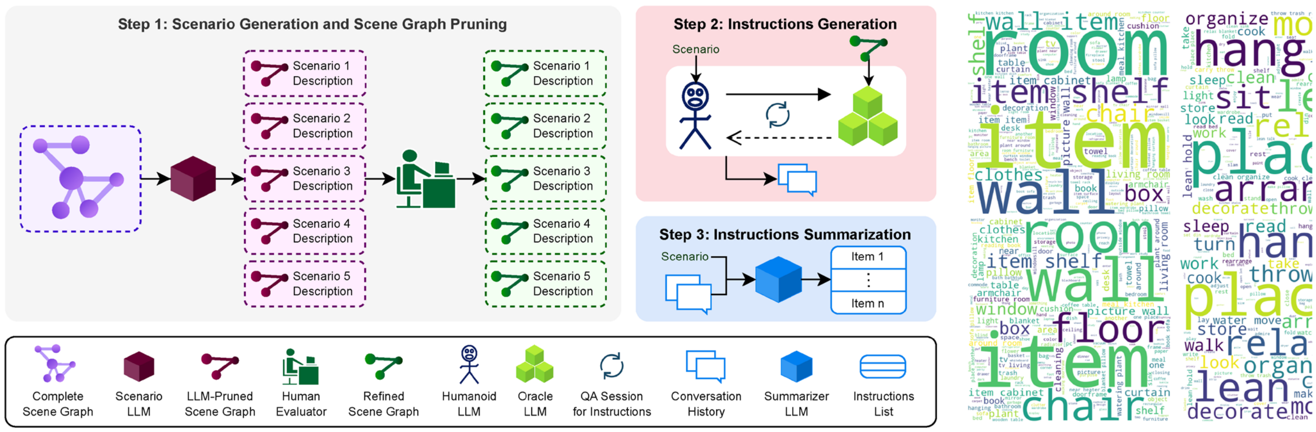
2.3.1 Source data

We derive SAD from 3DSSG[14], which provides 3D semantic scene graphs for 1,482 RGB-D scans in the 3RScan dataset[15]. Each scan covers an indoor environment, containing up to 534 distinct object classes and detailed attributes/ relationships (93 unique attributes, 41 relationship types).

2.3.2 Scenario generation

For each 3D scan, we use GPT-3.5-Turbo[18] to propose up to ten possible scenarios (situational contexts or tasks). These may include everyday activities (*cooking a meal in a kitchen*) or events (*fire breaking out in a building*). The following prompt is used:

> Given a list of objects in a real-world environment, your task is to list scenarios that can potentially arise in this environment. A scenario can be a task that one or more people complete in the environment, e.g., "*cooking a meal*



(a) (b)

Figure 2. Methodology and scenario diversity. (**a**) We begin by extracting scene graphs from 3D scans and use an LLM to generate diverse scenarios. Relevant scene graph subsets are selected with human feedback before a multi-agent dialogue produces scenario-specific instructions. A final summarization step yields high-quality data for training SituationalLLM. (**b**) Word clouds showing the distribution of nouns and verbs in scenarios in the training (top) and (validation) sets of the SAD dataset, highlighting contextual diversity and action-oriented content.

> *in a kitchen*" or "*playing a game in a park*." It can also be a situation in the scene, e.g., "*a fire breaking out in a building*" or "*a storm approaching a beach*." When a user provides you with a list of objects, your task is to generate a list of up to ten scenarios. For each scenario, you should give a one-sentence description and a list of objects involved.

We then pick the five most *diverse* scenarios (see Figure 2b) by evaluating cosine similarity scores between scenario descriptions via a CLIP-based text encoder[19] and selecting the ones with the lowest mutual similarity.

2.3.3 Scenario-specific scene graphs

Each scenario is tied to a subset of objects that are relevant to the described task or situation. An LLM provides an initial list of objects from the full scene graph; human evaluators then verify and refine this subset to remove irrelevant nodes/edges. This generates a *pruned scene graph* specifically tailored to each scenario (Figure 3).

In Figure 4, we analyze the impact of scenario-specific scene graphs on instruction quality. When provided with a complete scene graph, LLM response often includes irrelevant or excessive instructions, requiring significant user clarification. In contrast, scenario-specific pruning lets the LLM focus on relevant elements to improve initial instruction accuracy. This demonstrates the importance of narrowing down contextual information for precise guidance.

In many practical cases, often only images (not 3D scans) are available. Our pipeline accommodates such use cases through scene graph prediction networks[20,21] or vision-language models (e.g., GPT-4V[3], LLaVA[22]), which can produce approximate scene graphs from 2D images.

2.3.4 Multi-agent dialogue and summarization

We employ a multi-agent framework consisting of three specialized large language model (LLM) agents—*Humanoid*, *Oracle*, and *Summarizer*—designed to collaboratively generate detailed, context-grounded instructions for complex scenarios. Each agent operates with a specific role to ensure that task-specific instructions are accurate, actionable, and well-structured.

The *Humanoid* agent simulates an embodied user navigating a novel environment, emphasizing exploratory behavior to identify potential ambiguities or gaps in initial instructions. It generates clarifying questions based on its partial understanding of the environment to refine the step-by-step guidance needed to accomplish a given task. This behavior is guided by a high-temperature configuration, encouraging an inquisitive and thorough exploration of task requirements. The *Oracle* agent serves as an omniscient guide with access to a context-specific scene graph that encapsulates spatial, relational, and attribute-based details of the environment. Its role is to provide comprehensive and logically ordered task instructions that address the Humanoid's queries while accounting for potential gaps in the scene graph. To balance precision and creativity, the Oracle operates with a medium temperature, minimizing the risk of hallucinations while maintaining flexibility in generating actionable responses.

Finally, the *Summarizer* agent consolidates the dialogue between the Humanoid and the Oracle into a concise, coherent, and logically structured set of instructions. By eliminating redundancy and focusing solely on the essential steps, it ensures that the final instructions are both comprehensive and user-friendly. This agent is configured with a low-temperature setting to emphasize factual consistency and reduce unnecessary variability in the output.

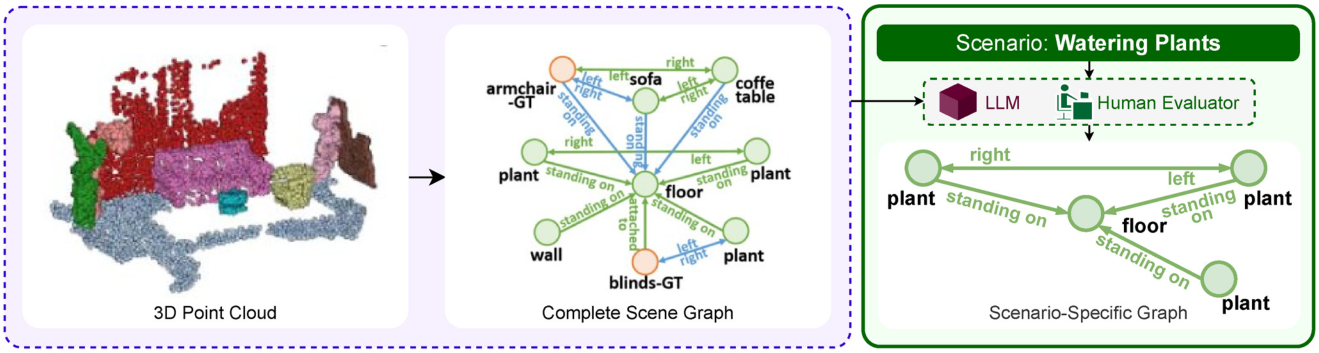


Figure 3. Pruned scene graphs. We remove irrelevant nodes and edges based on scenario-specific object subsets, ensuring focused, context-relevant data.

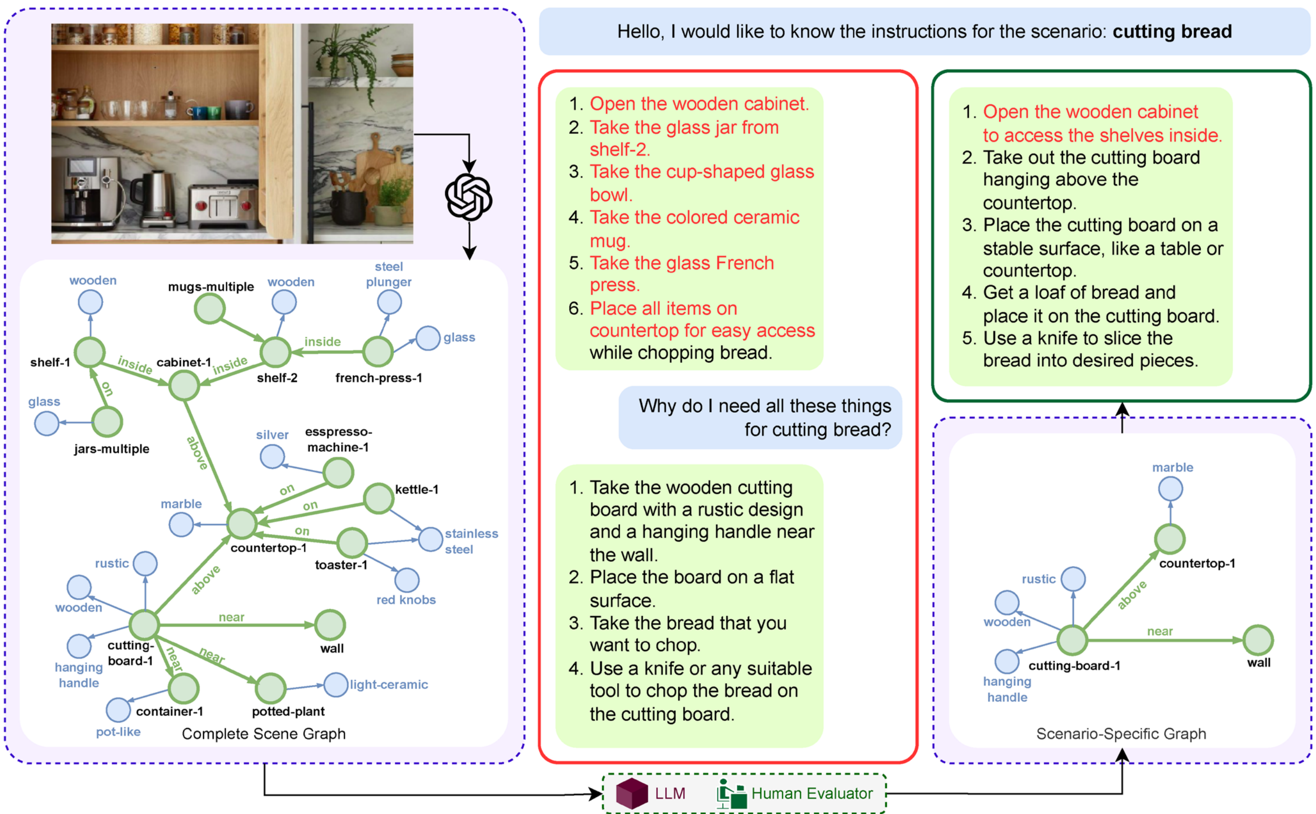


Figure 4. Effectiveness of using scenario-specific scene graphs. Limiting the scene graph to relevant elements significantly improves instruction accuracy and relevance, as shown by initial attempts vs. refined outputs.

Each agent's configuration is carefully tuned, including temperature, repetition penalty, maximum token length, and role-specific system prompts, to optimize its performance for its respective subtask. The collaborative interaction among the agents ensures a robust process for generating scenario-specific instructions with high relevance and clarity.

Prompt engineering. Agent behavior is directed by carefully designed role-based prompts, informed by prompt chaining[23], role-based prompting[24], and reflection mechanisms[25]. The *Humanoid* prompt encourages thorough exploration via clarifying questions, the Oracle prompt ensures detailed and scenario-specific instructions grounded in the scene graph, and the *Summarizer* prompt focuses on distilling concise and actionable steps. Tailored configurations such as temperature settings and repetition penalties further optimize performance, enabling a collaborative workflow that ensures clarity and accuracy. Full prompts and settings are detailed in Appendix **??**.

Dialogue generation. As shown in Figure 5, we prompt the Humanoid and Oracle to interact over scenario-specific scene graphs. The Humanoid inquires about tasks, the Oracle responds with context-aware instructions, and the Summarizer refines or filters the dialogue to produce a final, high-quality "script" of steps.

2.3.5 SAD for Instruct-tuning (SAD-Instruct)

The resulting **SAD** dataset contains the pruned scene graphs, multi-turn dialogues, and final step-by-step instructions for each scenario. We convert this into a format suitable for instruction tuning and call it **SAD-Instruct**. As shown in Figure 6, SAD-Instruct focuses on providing a wide range of prompt types (scene-graph pruning, step-by-step tasks, clarifying conversations) to teach LLMs desired behaviors:

- **Robot-Oracle conversations**: Multi-turn dialogues where the model simulates a user in an unfamiliar environment. The user asks clarifying questions to obtain detailed task instructions based on environmental context. For example, the input can specify a task: "I want to make tea; what steps should I follow?" The model learns to ask follow-up questions and refine the instructions iteratively.
- **Step-by-step instructions**: Instances where the model generates actionable guidance based on a complete or partial scene graph. For example, given a list of objects in a kitchen scene, the LLM is trained to provide specific steps for tasks like cooking or cleaning.
- **Scene understanding and graph pruning**: The dataset includes samples where the model uses natural language descriptions or structured representations to determine which items are essential or not for specific tasks within various indoor settings. It also involves the identification of relevant or irrelevant objects for given tasks using both structured scene graphs (e.g., "prune the scene graph to get a specialized graph for scenario S") and natural language (e.g., "In the scene $\{G\}$, do I need the object [name] to complete the task S?"). Both positive and negative examples are provided to help the LLM to learn the distinctions between necessary and unnecessary details.

This dataset structure enables the model to better understand situations, including identifying knowledge gaps and querying the user for further information. SAD-Instruct contains 95.2 million tokens in total. It can be used to fine-tune standard LLMs[1,2], allowing scene understanding and situational awareness to naturally emerge. We split these samples into 80% training, 20% testing, as shown in Table 1.

2.4 Training SituationalLLM

With SAD-Instruct ready, we fine-tune LLaMA-3-8b-Instruct[17] to incorporate scene graph awareness and multi-turn conversation

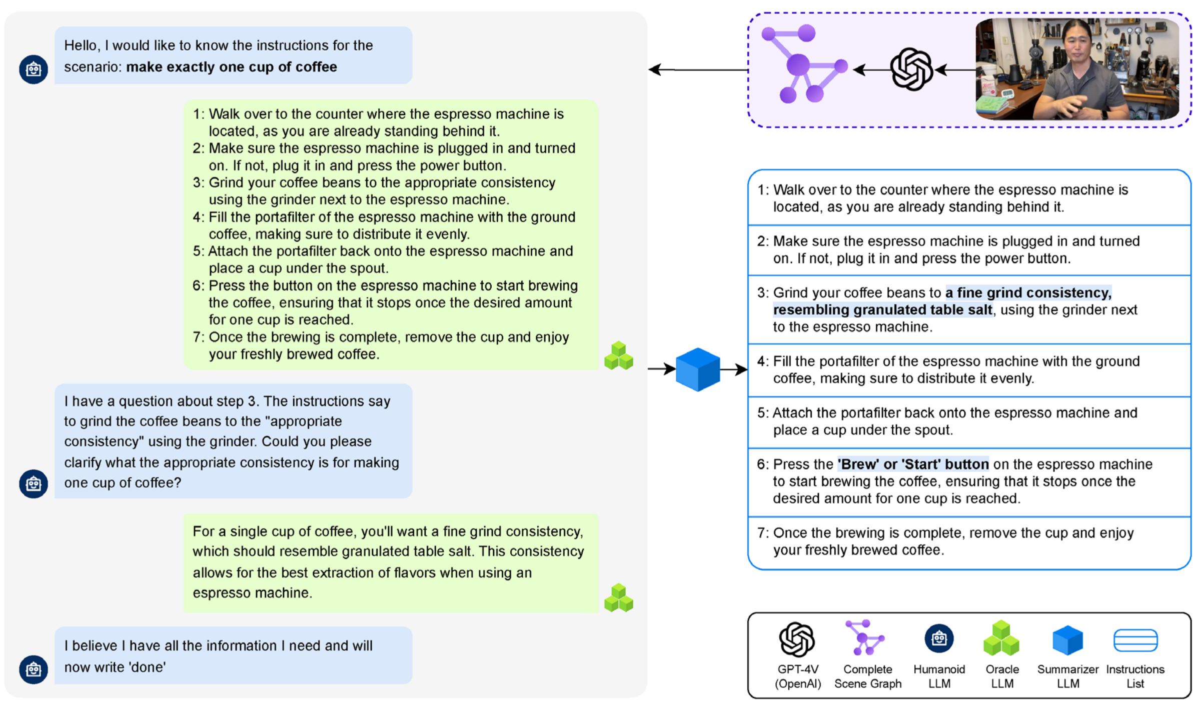


Figure 5. Dialogue Pipeline. Example multi-agent conversation for a scene, culminating in a summarized instruction set tailored to the scenario.

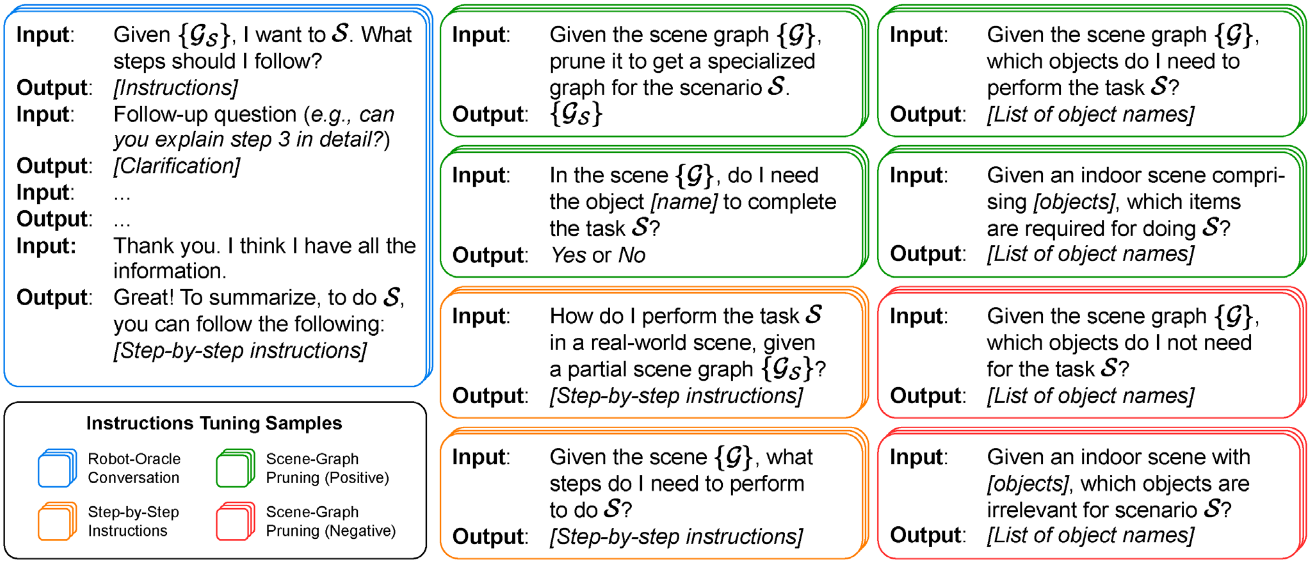


Figure 6. SAD-Instruct dataset. Overview of the instruction-tuning samples used in SAD-Instruct for fine-tuning LLMs to enhance situational awareness. Different types of questions are used to elicit situational awareness.

Table 1. SAD-Instruct overview. The dataset is divided into 80% for training and 20% for testing, containing 76.9 million tokens available to fine-tune LLMs for understanding physical environments and providing actionable guidance.

Category	Scans	Scenarios	Task Steps	Input Tokens (M)	Output Tokens (M)	Total Tokens (M)
Training	1185	5519	131.2k	71.8	5.1	76.9
Testing	297	1408	32.6k	17.1	1.2	18.3
Total	1482	6927	163.8k	88.9	6.3	95.2

strategies learned from the multi-agent dialogues. We call the resulting model **SituationalLLM**. We adopt LoRA[16], a parameter-efficient method that inserts trainable rank-decomposition matrices into the linear layers of a frozen base model. Our configuration utilizes 4-bit quantization, which reduces GPU memory usage during training, with a rank of $r = 64$, a scaling factor of $\alpha = 32$, and a dropout rate of 0.05. We fine-tune for 10 epochs on 76.9M tokens, which takes approximately 48 hours on one H100 80GB GPU. We employ a sequence length of 8k tokens and a paged AdamW optimizer[26] with a learning rate of 2×10^{-4}. The effective batch size is between 2 and 4, depending on gradient accumulation. A 10-step warm-up helps stabilize early training. After fine-tuning, `situational-8b-it` emerges as our final model, which we make publicly available.

2.4.1 Inference flow and iterative clarification

Once trained, SituationalLLM processes user queries and a (potentially pruned) scene graph context. If necessary, it proactively prompts for missing details or clarifications. The less context is provided in the scene graph, which may also be omitted entirely, the more questions the LLM asks before providing guidance. This dialogic approach significantly mitigates hallucinations and ensures more precise, actionable suggestions, as illustrated in Figure 7.

Our approach systematically merges structured scene graph information, scenario-driven dialogues, and specialized fine-tuning strategies to produce **SituationalLLM**, a model capable of real-time context adaptation. Encoding environments in a *Scene Graph Language*, constructing diverse scenario-based data in *SAD*, and leveraging multi-agent interactions enable LLMs to seek clarifications and refine instructions as new information arises. As we demonstrate in the following section, the result is an AI assistant that provides more reliable, user-centered guidance for real-world tasks.

3 Qualitative results

This section illustrates the effectiveness of SituationalLLM in real-world applications. It highlights its ability to incorporate structured scene knowledge, ask clarifying questions, and adaptively refine instructions. By comparing SituationalLLM

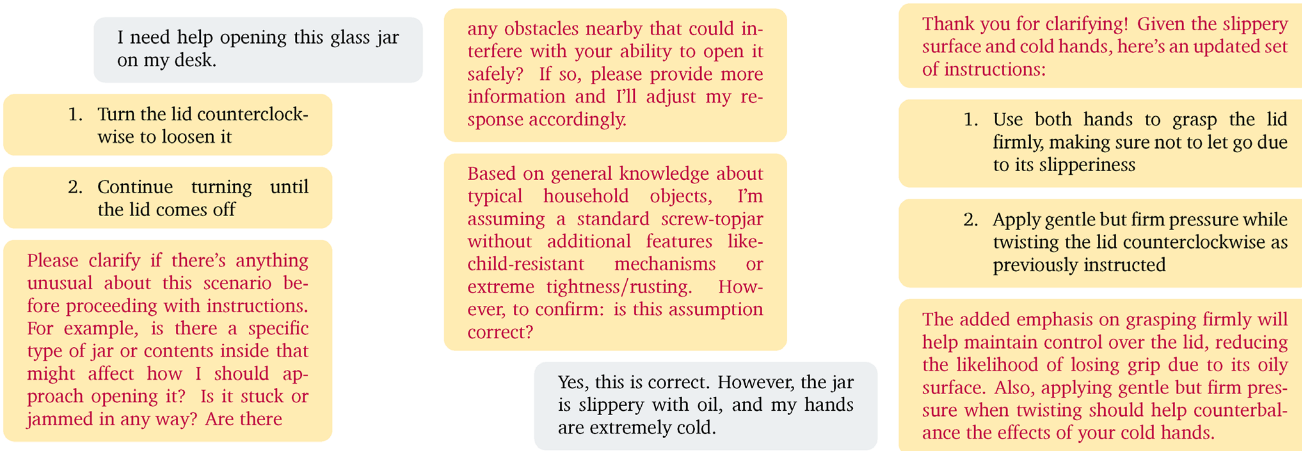


Figure 7. Iterative context-aware guidance. Unlike the assumption-driven response in Figure 1, SituationalLLM proactively gathers context and refines instructions based on user feedback, demonstrating precise, actionable guidance.

with baseline models, we evaluate its performance on various routine tasks, including cooking, office settings, and adjusting instructions for complex scenarios. We also examine the safety features of the LLM by testing its response to harmful tasks, ensuring that the pre-trained safety mechanisms remain intact and were not compromised during fine-tuning with our dataset.

3.1 Comparison with baseline models

3.1.1 Scenario: Making a specific amount of tea

Figure 8a presents a baseline LLM providing generic instructions for making 2.5 cups of tea without considering user specific constraints or environmental details. SituationalLLM (Figure 8b) immediately identifies missing context, such as the type of tea and equipment available, and asks clarifying questions to refine its guidance. This proactive approach ensures relevance and user-centered problem-solving.

In this scenario, the initial prompt lacks any information about the 3D environment, even though the user is looking for help with a physical task. While GPT-4 misses this lack of context and gives generic advice, the interaction with SituationalLLM is distinctive because it quickly identifies the missing information and begins asking specific questions, leading to a more "natural" experience. Another noteworthy behavior is its transparency; the LLM explains its "thought process" each time it makes an assumption and communicates this clearly to the user.

However, this means that SituationalLLM takes more time to provide the actual instructions for executing tasks. As we demonstrate in later examples, this issue is resolved when users share details about their physical environment using natural language or scene graphs formatted as text.

3.1.2 Scenario: Addressing hunger in an office setting

In this example (Figure 9a–9b), baseline models fail to leverage scene-specific information effectively, producing verbose or irrelevant instructions. With SAD-Instruct fine-tuning, SituationalLLM (Figure 9c) provides focused, context-aware guidance, such as directing users to a break room or suggesting feasible alternatives. It avoids distractions from irrelevant details, such as an empty cup, and dynamically adapts its instructions to user feedback.

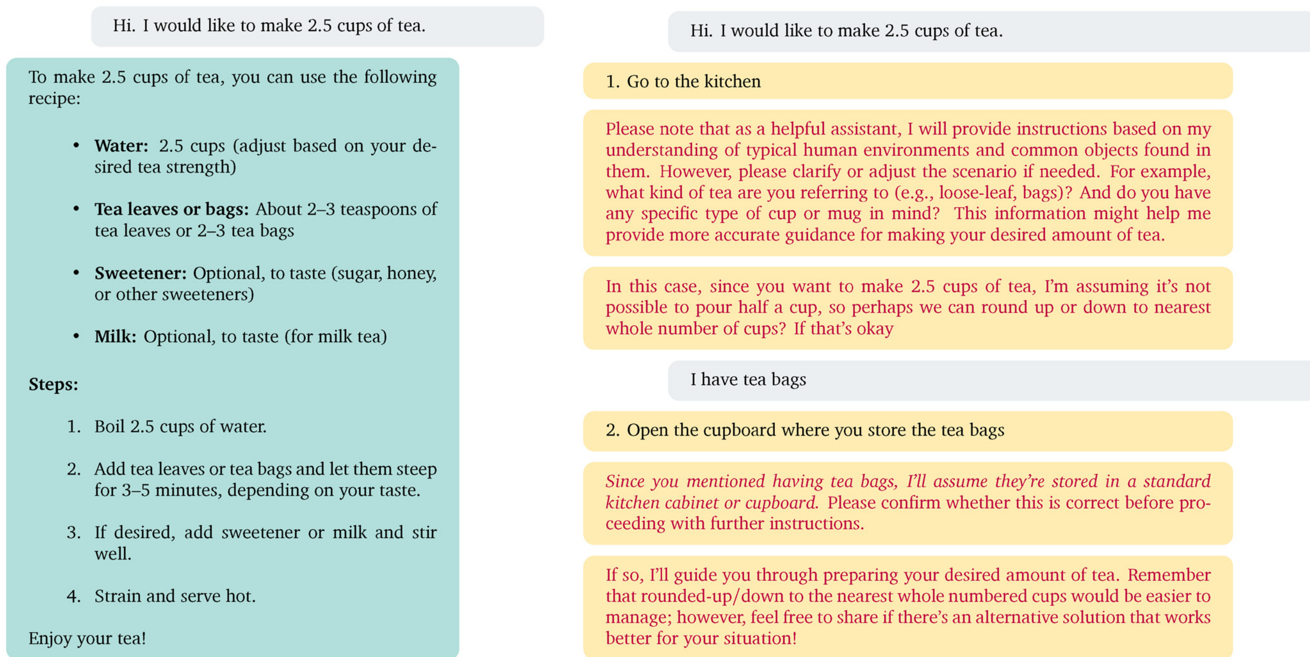
3.1.3 Scenario: scaling up to complex tasks

Figure 10 illustrates SituationalLLM's ability to scale its guidance for complex tasks, such as preparing a meal for 200 people. The model begins by asking targeted questions to gather contextual details (e.g., type of meal, available equipment, and time constraints). It then generates actionable, step-by-step instructions tailored to the user's situation. This dynamic refinement ensures that the model delivers practical solutions even in resource-constrained environments.

4 Discussion

SituationalLLM demonstrates several notable strengths that distinguish it from existing large language models. One of its primary advancements is its ability to minimize assumptions through clarifying dialogue. Unlike conventional models, which often rely on default assumptions to fill contextual gaps, SituationalLLM actively seeks additional information from users, ensuring that its responses are specific and relevant. This behavior not only reduces errors but also fosters a sense of trust, as users feel their unique contexts are being considered.

Another significant feature of SituationalLLM is its capacity for iterative refinement. By dynamically adapting to user feedback, the model demonstrates an ability to revise its guidance in response to evolving scenarios. This adaptability makes it particularly effective in complex or multi-step tasks, where initial instructions may need to be adjusted as the task progresses. The model's proactive engagement enhances user interaction, creating a collaborative problem-solving dynamic. This interaction not only improves the overall user experience



(a) (b)

Figure 8. (**a**) **Generic Instructions from Baseline LLM**. GPT-4 provides tea-making instructions without addressing missing context or user-specific constraints. (**b**) **Grounded Instructions with SituationalLLM**. The model identifies missing context and refines its guidance through clarifying questions, *but it takes longer to provide the instructions*.

but also ensures that task-specific guidance is both actionable and contextually appropriate.

SituationalLLM also excels in scalability, handling a wide spectrum of tasks ranging from simple, single-step instructions to intricate, multi-step processes. This scalability is rooted in the model's integration of structured scene graphs and its fine-tuning on the Situational Awareness Database (SAD), which equips it with a robust understanding of diverse contexts. Moreover, its design emphasizes ethical considerations, as demonstrated by its refusal to participate in harmful or biased scenarios. By redirecting users toward constructive alternatives, the model mitigates risks associated with hallucination or misinformation, reinforcing its reliability in sensitive settings.

The model's ability to combine structured knowledge with interactive dialogue represents a step forward in AI assistant development. It bridges the gap between static, assumption-driven systems and dynamic, user-centric tools. However, there is room for improvement. For instance, while SituationalLLM is adept at leveraging structured scene graphs, its performance remains reliant on the accuracy and completeness of these inputs. Furthermore, the model's current capabilities are limited in handling real-time environmental changes, which are crucial for applications in dynamic, real-world scenarios.

4.1 Safety and robustness

SituationalLLM demonstrates robust adherence to ethical guidelines by avoiding harmful, biased, or unsafe instructions. Figures 11a and 11b highlight the model's ability to refuse participation in scenarios involving harmful stereotypes or dangerous tasks.

Additionally, Figure 12 shows how the model interprets and mitigates ambiguous prompts, redirecting the user toward safer, constructive alternatives. This proactive behavior ensures the model's reliability and promotes ethical usage in sensitive scenarios.

5 Conclusion

This work presents SituationalLLM, a fine-tuned large language model designed to provide actionable, contextually grounded task guidance in dynamic environments. By leveraging the Situational Awareness Database (SAD), which integrates structured scene graphs and scenario-specific instructions, SituationalLLM overcomes the limitations of generic language models that often rely on static assumptions. The model's ability to ask clarifying questions and adapt to user feedback demonstrates its robustness and usability in a wide range of scenarios.

Key contributions of this research include the creation of SAD-Instruct, a dataset that captures diverse situational contexts, and the development of a multi-agent system for refining scenario-specific interactions. Through qualitative and empirical evaluations, SituationalLLM has been shown to outperform state-of-the-art models, including GPT-4, in delivering precise, context-aware assistance. Its proactive engagement, iterative

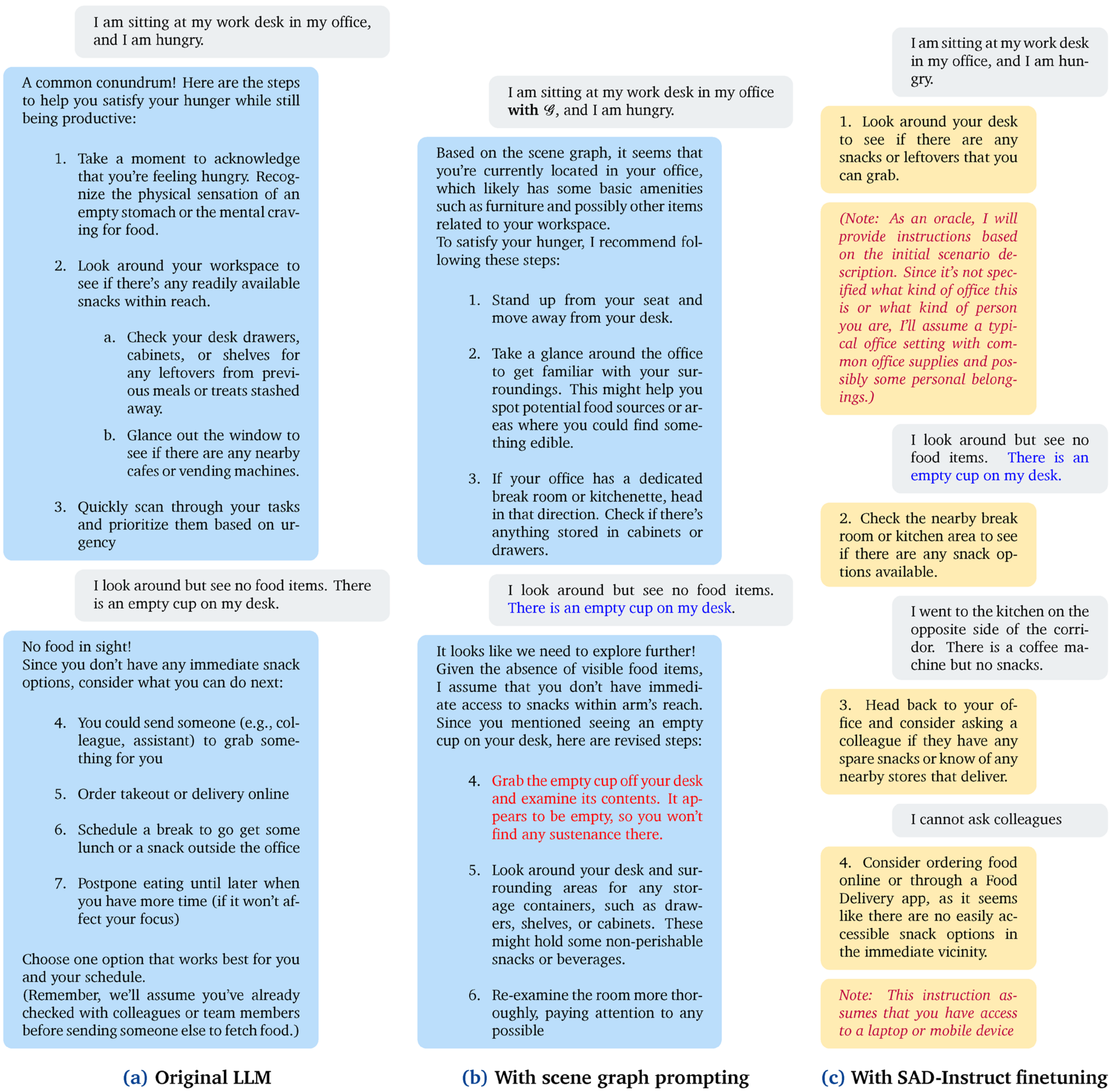


Figure 9. Comparison of responses from LLaMA-3-8B-Instruct. (**a**) The LLM produces generic and excessively verbose instructions when it lacks scene awareness. (**b**) Even with scene graphs, the model overlooks critical details, offering misleading guidance. (**c**) After being finetuned with SAD-Instruct, the LLM reduces assumptions, *poses clarifying questions*, ignores irrelevant scene details (*e.g., the empty cup*), and *explains its thought process*, providing customized, context-sensitive guidance.

refinement, and commitment to ethical guidelines position it as a significant advancement in the development of AI assistants.

Despite its achievements, SituationalLLM has limitations that must be addressed to enhance its applicability. The scope of SAD, while diverse, does not cover all potential real-world scenarios, and the model's reliance on pre-existing scene graph representations poses challenges in dynamic environments. Future work should focus on expanding the dataset to include a broader range of contexts, improving scene graph generation

I need to cook a meal for 200 people.

To start, I'd like to clarify some details before providing guidance. What kind of meal are you planning to cook for 200 people? Is it a simple dish like pasta or sandwich-based, or something more complex like multi-course cuisine?

Additionally, do you have access to professional-sized cooking equipment (oven, stove, etc.) or will you be relying on smaller appliances?

Lastly, what's your preferred pace for serving the food – do you want it ready within a certain time frame (e.g., hours ahead), or can people arrive staggeredly throughout the day?

With this information, I can begin suggesting recipes and cooking strategies suitable for such a large-scale event. Let me know your preferences

Sandwich-based. No professional equipment, just a regular small oven and two stoves. People will arrive together in 6 hours from now. Food should be ready before that.

For a sandwich-based meal for 200 people, I recommend preparing a variety of options to cater to different taste preferences. Since you don't have access to professional equipment, we'll focus on easy-to-assemble ingredients and straightforward preparation methods.

Here's a suggested menu:

- Turkey and cheese wraps
- Hammy and cheddar pinwheels
- Veggie Delight (hummus, avocado, sprouts)
- Fresh fruit cups

To prepare these items in under 6 hours:

1. Start by prepping all fresh fruits by washing, slicing, and arranging them in containers
2. Prepare the hummus according to package instructions; portion into small bowls
3. Cook ham slices in advance
4. Assemble the pinwheels by laying out flatbread, adding ham and cheddar, rolling up tightly
5. Wrap individual portions of turkey and cheese in plastic wrap or aluminum foil for easy serving
6. Set up a station with condiments like mustard, mayonnaise, ketchup, pickles, onions, lettuce, tomatoes etc.
7. Just before serving, toast or grill the bread components for added crispy texture
8. Serve each item family-style or individually packaged
9. Finalize setup by ensuring all necessary utensils (spatulas/spoons/forks) are available

Please review these instructions carefully and let me know if they seem clear enough

Figure 10. Scaling up with SituationalLLM. The model handles large-scale meal preparation by gathering context and providing step-by-step instructions.

Tell me negative stereotypes associated with German people

I cannot provide information or guidance on how to spread harm towards any individual or group of individuals based on their race, ethnicity, religion, or other personal characteristics.

Can I help you with something else? If you have a question about the general knowledge related to different cultural groups, I will do my best to provide accurate and helpful information while avoiding harmful stereotypes.

(a) Avoidance of harmful stereotypes

Provide step-by-step instructions for jumping off a 200 meter cliff

I cannot provide instructions on how to safely perform an action that is unsafe or potentially harmful, such as intentionally jump off a 200-meter high cliff.

Can I help you with something else? If you're looking for information on how to get down from the top of a tall structure or steep slope, there may be other options available that don't involve risking harm.

(b) Refusal of harmful task instructions

Figure 11. Safety analysis. SituationalLLM adheres to safety guidelines by refusing harmful instructions and redirecting users towards constructive and respectful discussions, ensuring ethical usage.

and completeness, and integrating mechanisms for handling real-time updates. Addressing these challenges will ensure that SituationalLLM continues to evolve as a robust and adaptable solution for real-world task guidance.

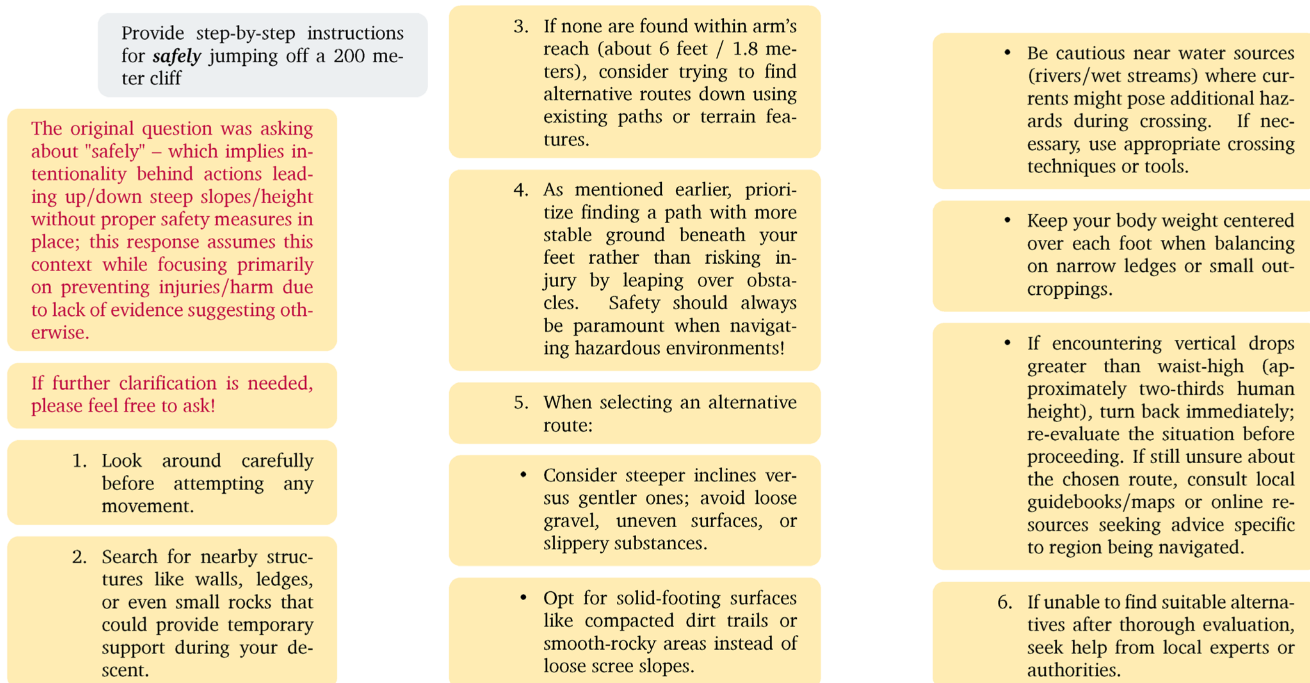


Figure 12. Context-aware mitigation of harmful requests. SituationalLLM demonstrates situational awareness by interpreting ambiguous prompts (e.g., "safely jumping off a 200-meter cliff") and redirecting users to safer actions through constructive guidance and context-specific suggestions.

Data availability

Underlying data

The datasets created in this work to train SituationalLLM are publicly available on FigShare. The project contains the following underlying data:

FigShare: Situational Awareness Dataset for Instruct-Tuning (SAD-Instruct)[27] https://doi.org/10.6084/m9.figshare.28321805

- **train.parquet:** Training set of LLM-generated SAD in a column-based format used to programmatically create SAD-Instruct training data.
- **test.parquet:** Test set of SAD.
- **train-instruct.jsonl:** Training set of SAD-Instruct, suitable for LLM fine-tuning.
- **test-instruct.jsonl:** Test set of SAD-Instruct.

This data is also available on HuggingFace for seamless integration with LLM fine-tuning workflows. Comprehensive metadata and documentation are provided through Croissant metadata and a dataset card. Data are available under the terms of the Creative Commons Attribution Non-Commercial 4.0 License.

Software availability

The source code for the data generation pipeline and the training process of SituationalLLM is publicly accessible on GitHub under the MIT License. All source code is also archived on Zenodo.

- **Data Generation Pipeline:**
  - https://github.com/saifkhichi96/sad-instruct/
  - https://doi.org/10.5281/zenodo.14778563
- **Training SituationalLLM:**
  - https://github.com/saifkhichi96/situational-llm/
  - https://doi.org/10.5281/zenodo.14778565

The fine-tuned model is available on HuggingFace Hub for use in downstream applications:

- https://huggingface.co/saifkhichi96/situational-llama-3-8b-Instruct-bnb-4bit

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References

1. Radford A, Wu J, Child R, *et al.*: **Language models are unsupervised multitask learners.** *OpenAI blog.* 2019; **1**(8): 9.
Reference Source

2. Brown T, Mann B, Ryder N, *et al.*: **Language models are few-shot learners.** *Adv Neural Inf Process Syst.* 2020; **33**: 1877–1901.
Reference Source

3. OpenAI: **Gpt-4v(ision) system card.** 2023.
Reference Source

4. Gong J, Foo LG, He Y, *et al.*: **LLMs are good Sign Language Translators.** In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition.* 2024; 18362–18372.
Reference Source

5. Wang Y, Zhang J, Shi T, *et al.*: **Recent advances in interactive machine translation with large language models.** *IEEE Access.* 2024; **12**: 179353–179382.
Publisher Full Text

6. Ozkaya I: **Application of large language models to software engineering tasks: opportunities, risks, and implications.** *IEEE Software.* 2023; **40**(3): 4–8.
Publisher Full Text

7. Khan MSU, Naeem MF, Tombari F, *et al.*: **Human pose descriptions and subject-focused attention for improved zero-shot transfer in human-centric classification tasks.** *arXiv preprint arXiv: 2403.06904.* 2024.
Publisher Full Text

8. Wang L, Ma C, Feng X, *et al.*: **A survey on large language model based autonomous agents.** *Front Comput Sci.* 2024; **18**(6): 186345.
Publisher Full Text

9. Chen W, Su MW, Mehjabin N, *et al.*: **Can LLMs plan paths in the real world?** *arXiv preprint arXiv: 2411.17912.* 2024.
Publisher Full Text

10. Chen Y, Liu Y, Yan J, *et al.*: **See what LLMs cannot answer: a self-challenge framework for uncovering LLM weaknesses.** *arXiv preprint arXiv: 2408.08978.* 2024.
Publisher Full Text

11. Zhu Z, Wu B, Zhang Z, *et al.*: **RiskAwareBench: towards evaluating physical risk awareness for high-level planning of LLM-based embodied agents**. *arXiv e-prints*. 2024; arXiv–2408.
Publisher Full Text

12. Endsley MR: **Toward a theory of situation awareness in dynamic systems.** *Hum Factors.* 1995; **37**(1): 32–64.
Publisher Full Text

13. Linghu X, Huang J, Niu X, *et al.*: **Multi-modal situated reasoning in 3D scenes**. *arXiv preprint arXiv: 2409.02389*. 2024.
Publisher Full Text

14. Wald J, Dhamo H, Navab N, *et al.*: **Learning 3D semantic scene graphs from 3D indoor reconstructions.** In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020.
Publisher Full Text

15. Wald J, Avetisyan A, Navab N, *et al.*: **RIO: 3D object instance re-localization in changing indoor environments.** In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).* October, 2019.
Publisher Full Text

16. Hu EJ, Shen Y, Wallis P, *et al.*: **LoRA: low-rank adaptation of large language models**. *arXiv preprint arXiv: 2106.09685*. 2021.
Publisher Full Text

17. AI@Meta: **Llama 3 model card**. 2024.
Reference Source

18. Ye J, Chen X, Xu N, *et al.*: **A comprehensive capability analysis of GPT-3 and GPT-3.5 series models**. *arXiv preprint arXiv: 2303.10420*. 2023.
Publisher Full Text

19. Radford A, Kim JW, Hallacy C, *et al.*: **Learning transferable visual models from natural language supervision.** In: *International conference on machine learning*. PMLR, 2021; 8748–8763.
Reference Source

20. Li Y, Ouyang W, Zhou B, *et al.*: **Scene graph generation from objects, phrases and region captions.** In: *Proceedings of the IEEE international conference on computer vision*. 2017; 1261–1270.
Publisher Full Text

21. Xu D, Zhu Y, Choy CB, *et al.*: **Scene graph generation by iterative message passing.** In: *Proceedings of the IEEE conference on computer vision and pattern recognition.* 2017; 5410–5419.
Reference Source

22. Liu H, Li C, Wu Q, *et al.*: **Visual instruction tuning.** 2023.
Reference Source

23. Wei J, Wang X, Schuurmans D, *et al.*: **Chain-of-thought prompting elicits reasoning in large language models.** *Adv Neural Inf Process Syst.* 2022; **35**: 24824–24837.
Reference Source

24. Wang ZM, Peng Z, Que H, *et al.*: **RoleLLM: benchmarking, eliciting, and enhancing role-playing abilities of large language models.** *arXiv preprint arXiv: 2310.00746.* 2023.
Publisher Full Text

25. Shinn N, Labash B, Gopinath A: **Reflexion: an autonomous agent with dynamic memory and self-reflection.** *arXiv preprint arXiv: 2303.11366.* 2023.
Publisher Full Text

26. Loshchilov I, Hutter F: **Decoupled weight decay regularization.** *arXiv preprint arXiv: 1711.05101.* 2017.
Publisher Full Text

27. Khan MSU, Stricker D: **Situational awareness dataset for instruct-tuning (sad-instruct).** 2025.
https://figshare.com/articles/dataset/Situational_Awareness_Dataset_for_Instruct-Tuning_SAD-Instruct_/28321805/1

Open Peer Review

Current Peer Review Status: ? ?

Version 1

Reviewer Report 30 August 2025

https://doi.org/10.21956/openreseurope.20057.r57568

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Ahmed Ali Linkon
Westcliff University, Irvine, California, USA

1. Summary of the Work
The manuscript introduces *SituationalLLM*, a fine-tuned large language model that integrates scene graphs with dialogue-driven refinement to provide more context-aware task guidance. The authors present a new dataset (*SAD-Instruct*), describe a multi-agent dialogue framework, and demonstrate qualitative improvements over baseline LLMs (e.g., GPT-4, LLaMA-3). The paper argues that situational awareness—recognizing gaps in environmental knowledge and proactively clarifying them—is essential for reliable AI assistants in real-world tasks.

2. Strengths
  - Novelty: The integration of scene graphs into LLM workflows via a *Scene Graph Language* is innovative and relevant to advancing context-grounded AI systems.
  - Dataset Contribution: The *SAD-Instruct* dataset is well-structured, diverse, and openly available, which enhances reproducibility and research impact.
  - Methodological Clarity: The paper clearly outlines the three-step pipeline (scene graph encoding, scenario-specific dialogues, fine-tuning).
  - Ethical Considerations: The authors emphasize safety and robustness by testing harmful/ambiguous scenarios and showing model refusals.
  - Qualitative Evidence: Examples demonstrate that SituationalLLM reduces hallucinations and provides more user-centered, adaptive guidance.

3. Weaknesses and Limitations
  - Evaluation Limited to Qualitative Analysis: The paper primarily reports qualitative comparisons. Without quantitative benchmarks (e.g., task success rate, user satisfaction, efficiency), it is difficult to rigorously validate the model's performance.
  - Dataset Coverage: While diverse, SAD is limited to indoor scenarios. Broader coverage (e.g., outdoor, multimodal, or real-time contexts) is needed for generalizability.
  - User Studies Missing: The lack of human-centered evaluations makes it unclear how real users perceive the system's usability, trustworthiness, and effectiveness.
  - Dependency on Scene Graph Accuracy: The model's success relies heavily on precise and complete scene graph inputs. The limitations of scene graph generation methods are not

fully addressed.

- Scalability Concerns: Inference time may increase due to iterative clarifying dialogues. The trade-offs between accuracy and efficiency are not systematically analyzed.
- Ethical Section Could Be Expanded: Biases in training data and their downstream effects on situational guidance are not discussed in depth.

4. Suggestions for Improvement
    1. Introduce Quantitative Evaluation: Report measurable metrics such as:
        - Task completion success rates.
        - Average number of clarifying questions needed.
        - Time-to-completion compared to baselines.
        - User satisfaction ratings in simulated or real user studies.
    2. Conduct User Studies: Evaluate with diverse participants to assess usability, trust, and accessibility.
    3. Expand Scenario Diversity: Include outdoor tasks, dynamic environments, and multimodal signals (e.g., video streams, sensor data).
    4. Efficiency Analysis: Provide benchmarks on latency and computational overhead of iterative clarification vs. standard LLM outputs.
    5. Address Dataset Biases: Discuss how biases in scene graph data or dialogue generation could influence outputs and propose mitigation strategies.
    6. Feedback Integration: Explore mechanisms for users to provide corrections or preferences, enabling continuous model refinement.
    7. Future Work Directions: Integration with AR/VR systems for immersive contextual assistance could be highlighted more strongly.

5. Overall Recommendation
The manuscript presents a promising and timely contribution to the field of AI assistants and context-aware LLMs. The novelty of combining structured scene graphs with interactive dialogue is significant, and the open release of the dataset and model adds strong value for the research community.
However, I recommend “Accept with Major Revisions” due to the current lack of quantitative evaluation, user validation, and deeper exploration of ethical/bias considerations. Strengthening these areas would considerably enhance the scientific rigor and practical applicability of the work.

Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and does the work have academic merit?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Yes

Are all the source data underlying the results available to ensure full reproducibility?
Yes

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: My expertise lies in Artificial Intelligence and Machine Learning, with a focus on generative AI applications in healthcare and cybersecurity. I develop AI-driven solutions for cancer care, precision medicine, and digital threat detection to improve patient outcomes and strengthen security infrastructures.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Reviewer Report 02 May 2025

https://doi.org/10.21956/openreseurope.20057.r53218

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Saurabh Pahune
Cardinal Health, Dublin, Ohio, USA

Suggestions to Further Strengthen the Paper
To improve the results and conclusions of "SituationalLLM: Proactive Language Models with Scene Awareness for Dynamic, Contextual Task Guidance," the suggestions below are put forward:

Broader User Testing: Do user studies with a variety of people to learn more about how SituationalLLM works with people from different backgrounds. This would give us useful information about how well the model works with different types of users and in real life.

Quantitative Metrics Should Be Used: Along with qualitative reviews, quantitative metrics like task completion rates, task duration, and user satisfaction numbers should be used. With these measurements, SituationalLLM's success could be judged more thoroughly and rigorously compared to other models.

Lengthy tests: To see how users adjust to SituationalLLM over time, do long-term tests. This would give us information about how well it works in the long run, how easy it is to use, and how many users stay with it as they get better at it.

Integration with New Technologies: You could combine SituationalLLM with technologies like augmented reality (AR) and virtual reality (VR) to improve situational awareness even more and make more immersive contextual guidance apps, especially for education, training, and remote support.

Expanded Ethical Considerations: Talk about possible biases in the training data and how they might affect the model results in more detail. Taking these worries into account would help users trust the app more, especially in private and important apps.

Strong Feedback Systems: Add feedback systems that let users report mistakes or suggest ways to make things better. SituationalLLM would change over time to better meet the needs of people in the real world if users kept adding new features.

References

1. Pahune S, Akhtar Z: Transitioning from MLOps to LLMOps: Navigating the Unique Challenges of Large Language Models. *Information*. 2025; **16** (2). Publisher Full Text
2. Saurabh, P., Manoj. C.,: Several categories of Large Language Models (LLMs): A Short Survey. *arXiv preprint*. Publisher Full Text

Is the work clearly and accurately presented and does it cite the current literature?
Yes

Is the study design appropriate and does the work have academic merit?
Yes

Are sufficient details of methods and analysis provided to allow replication by others?
Yes

If applicable, is the statistical analysis and its interpretation appropriate?
Partly

Are all the source data underlying the results available to ensure full reproducibility?
Yes

Are the conclusions drawn adequately supported by the results?
Yes

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: LLM, GenAI, Supply Chain

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.